

# Unlock You : Mathematical Optimization of Non-Recurring PIN Search

## Abstract :

Unlock You presents a mathematical framework for optimizing the search of fixed-length, non-recurring Personal Identification Numbers (PINs). Using permutation theory and probability analysis, the proposed model demonstrates how partial knowledge, such as the PIN length and known digits, significantly reduces the search space while improving the probability of identifying the correct PIN. A modified Back-Swap digit ordering algorithm is further introduced to prioritize user-biased digits, reducing the expected number of verification attempts without eliminating any valid candidate PINs. The derived mathematical relations are supported through a C-based implementation, serving as an educational demonstration of combinatorial optimization and search efficiency.

This provides a structured way to understand the geometry of the grid and the formation of winning patterns. The approach connects combinatorics and coordinate geometry to explain the game mathematically.

## Concepts Used :

Permutation and Probability Theory, Factorials,  
Combinatorics, Recursive Algorithms

## Introduction :

Brute-force search remains one of the simplest methods for verifying unknown PINs; however, its computational cost increases rapidly with the size of the search space. In practical situations, partial information such as the PIN length or one or more known digits can substantially reduce the number of possible candidates. This work investigates these reductions mathematically using permutation theory and probability analysis.

The proposed framework derives equations that quantify improvements in search performance and success probability under varying levels of prior knowledge. Additionally, a modified Back-Swap algorithm employing a user-biased shuffled digit array is introduced to improve the average search order while preserving complete coverage of all valid non-recurring PINs. The accompanying C implementation demonstrates the mathematical framework and illustrates how combinatorial techniques can be applied to optimize exhaustive search in an educational setting.

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# 1) Length of the PIN :

Knowledge of the user's PIN length (denoted by **N**) substantially reduces the number of possible PIN combinations. Consequently, the probability of correctly identifying the PIN increases, leading to improved algorithmic efficiency.

Let,

**Length of the PIN = N**

**Total possible digits = 10**

**Total sum of all permutations = 10!** (Instead of  $\sum_{n=1}^{10} \frac{10!}{(10-n)!}$  to ease calculation)

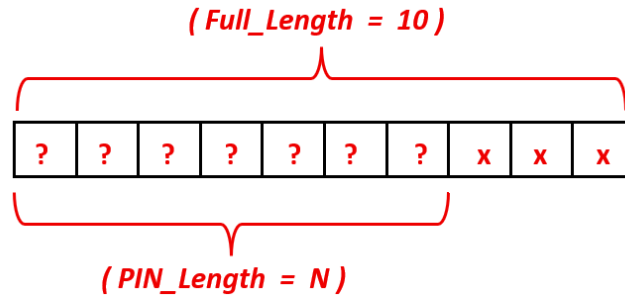


Figure 1: User PIN with known length.

Now,

$$\begin{aligned} \text{Probability of finding PIN without knowing PIN length} &= \frac{\text{P(Correct PIN) | Total no of PINs)}}{\text{n(Total sum of PINs)}} \\ &= \frac{\text{n(Correct PIN)}}{\text{n(Total sum of PINs)}} \end{aligned}$$

$$\therefore \text{Probability of finding PIN without knowing PIN length} = \left( \frac{1}{10!} \right) \dots\dots\dots \text{eq(i)}$$

Again,

$$\begin{aligned} \text{Probability of finding PIN with known PIN length} &= \frac{\text{P(Correct PIN) | Total no of PINs)}}{\text{n(Total PINs of known length)}} \\ &= \frac{\text{n(Correct PIN)}}{\text{n(Total PINs of known length)}} \\ &= \frac{1}{\frac{10!}{(10 - N)!}} \end{aligned}$$

$$\therefore \text{Probability of finding PIN with known PIN length} = \frac{(N - 1)!}{10!} \dots\dots\dots \text{eq(ii)}$$

Finally,

$$\begin{aligned} \text{Improvement ratio (I)} &= \frac{\text{P(Total PINs | Known Pin Length)}}{\text{P(Total PINs | Unknown Pin Length)}} \\ &= \frac{(10 - N)!}{\frac{10!}{10!}} \end{aligned}$$

$$\therefore \text{Improvement ratio (I)} = (N - 1)! \dots\dots\dots \text{eq(iii)}$$

When,

- PIN Length (**N = 4**), **Improvement ratio** =  $(10 - 4)! = 6! = 720\mathbf{x}$
- PIN Length (**N = 6**), **Improvement ratio** =  $(10 - 6)! = 4! = 24\mathbf{x}$
- PIN Length (**N = 8**), **Improvement ratio** =  $(10 - 8)! = 2! = 2\mathbf{x}$

## 2) No of known digits of the PIN :

Besides the PIN length, prior knowledge of  $n$  digits from a PIN of length  $N$  substantially decreases the number of possible PIN combinations. This reduction in the search space leads to a corresponding increase in the probability of successfully identifying the correct PIN.

Let,

**Length of the PIN =  $N$  ( $0 < N \leq 10$ )**

**No of known digits =  $n$  ( $0 < n \leq N$ )**

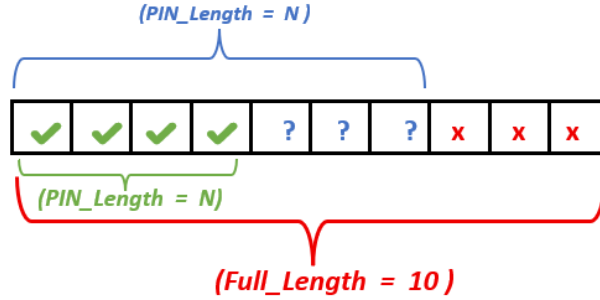


Figure 1: PINs with few no of known digits.

Now,

**Total PINs without known digits =  $P(10, N)$  ..... eq(i)**

Also,

**Total PINs with known digits =  $P(10 - n, N - n)$  ..... eq(ii)**

Then,

$$\begin{aligned} \text{Additional Increases in Chances (in \%)} &= \left\{ \frac{P(10, N)}{P(10 - n, N - n)} - 1 \right\} \times 100\% \\ &= \left\{ \frac{\frac{10!}{(10 - N)!}}{\frac{(10 - n)!}{\{10 - n - (N - n)\}!}} - 1 \right\} \times 100\% \\ &= \left\{ \frac{\frac{10!}{(10 - N)!}}{\frac{(10 - n)!}{\{10 - N\}!}} - 1 \right\} \times 100\% \end{aligned}$$

$$\therefore \text{Additional Increases in Chances (in \%)} = \left\{ \frac{10!}{(10 - n)!} - 1 \right\} \times 100\% \text{ ..... eq(iii)}$$

Again,

$$\begin{aligned} \text{When } (n = 1), \text{ Increases in Chances} &= \{10!/(10-1)! - 1\} * 100\% = 900\% \\ \text{When } (n = 2), \text{ Increases in Chances} &= \{10!/(10-2)! - 1\} * 100\% = 8900\% \\ \text{When } (n = 3), \text{ Increases in Chances} &= \{10!/(10-3)! - 1\} * 100\% = 71,900\% \end{aligned}$$

### 3) User Biased Shuffled digits Array:

**Let,**

Length of the PIN = N - 1

Length of Shuffled Array = 10 - 1 = 9

No of other likely correct digits = n

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 1 | 7 | 0 | 9 | 6 | 2 | 5 | 4 | 8 |
|---|---|---|---|---|---|---|---|---|---|

Figure 1: A User biased - shuffled digit Array.

**Now,**

**For Regular Array,**

$$\begin{aligned} \text{Total no of PINs} &= P(9, N - 1) \\ &= \frac{9!}{\{9 - (N - 1)\}!} \end{aligned}$$

$$\therefore \text{Total no of PINs} = \frac{9!}{(N - 1)!}$$

$$\therefore \text{Average no of Attempts (Regular)} = \frac{9!}{2(N - 1)!} \quad (1)$$

**Then,**

**For Biased - Shuffled Array,**

$$\begin{aligned} \text{Total no of PINs} &= P(9 - n, N - n - 1) \\ &= \frac{9!}{\{9 - n - (N - n - 1)\}!} \end{aligned}$$

$$\therefore \text{Total no of PINs} = \frac{(9 - n)!}{(10 - N)!} (1 - r) \quad \{r = \text{biased factor}\}$$

$$\therefore \text{Average no of Attempts (Regular)} = \frac{(9 - n)!(1 - r)}{2(N - 1)!} \quad (2)$$

**Again,**

$$\text{Increase in Chances (in \%)} = \left( \frac{\text{Attempt}_R}{\text{Attempt}_S} - 1 \right) \times 100\%$$

$$= \left\{ \frac{\frac{9!}{2 * (10 - N)!}}{\frac{(9 - n)!(1 - r)}{2 * (10 - N)!}} - 1 \right\} \times 100\%$$

$$\therefore \text{Increase in Chances (in \%)} = \left\{ \frac{9!}{(9 - n)!(1 - r)} - 1 \right\} \times 100\% \quad (3)$$

**In the above Eq(3),**

Since  $(1 < n < 9)$  and  $(0 < r < 1)$ , the denominator of Eq(3) is always positive.

Therefore, Eq(3) yields a positive percentage increase, mathematically confirming that prioritizing user-biased digits improves the expected search efficiency and increases the likelihood of identifying the correct PIN in fewer verification attempts.

#### 4) Total no of PINs and its Probabilitypercentage :

Let,

Total no of PINs with unknown length = 10!

Total no of PINs(When first digit is unknown) =  $\frac{10!}{(10 - N)!}$

Now

$$\begin{aligned}\text{Probability Percentage of finding the PIN with knowing the PIN length} &= \frac{n(\text{Correct PIN})}{n(\text{Total Possible PINs})!} * 100\% \\ &= \left( \frac{1}{10!} \right) * 100\% \\ &= \frac{(10 - N)!}{10!} * 100\% \\ \therefore \text{Probability Percentage of finding the PIN with knowing the PIN length} &= \frac{(10 - N)!}{36288} \% \quad (1)\end{aligned}$$

Again,

$$\begin{aligned}\text{Total no of PINs (after both first and length are known)} &= P(10 - 1, N - 1) \\ &= P(9, N - 1) \\ &= \frac{9!}{\{9 - (N - 1)\}!}\end{aligned}$$

$$\therefore \text{Total no of PINs with known length and fixed first digit} = \frac{9!}{(10 - N)!} \quad (2)$$

Also,

Let, the factorial value of (10 - N)! be **Fact - Value**.

Then,

From Eq(1),

$$\begin{aligned}\text{PINs probability percentage} &= \frac{(10 - N)!}{36,288} \\ &= \frac{\text{Fact - Value}}{36,288}\end{aligned}$$

$$\therefore \text{Fact - Value} = \text{PINs Probability Percentage} * 36,288 \quad (3)$$

Again,

From Eq(2),

$$\begin{aligned}\text{Total no of PINs (with length and first digit)} &= \frac{9!}{(10 - N)!} \\ &= \frac{9!}{\text{Fact - Value}} \\ \text{Total no of PINs} * \text{Fact - Value} &= 9! \\ \text{Total no of PINs} * (\text{PINs Probability Percent} * 36288) &= 9! \quad \text{From eq(3)} \\ \text{Total no of PINs} * \text{PINs Probability Percent} &= 10\end{aligned}$$

$$\therefore \text{Total no of PINs} = \left( \frac{10}{\text{PINs Probability}} \right) \quad (4)$$

$$\therefore \text{PINs Probaility Percentage} = \left( \frac{10}{\text{Total PINs}} \right) \quad (5)$$